

Journaling



Overview

- Write Image Journal (WIJ).
 - Processing.
 - Configuration.
 - Recovery.
- Journal (JRN).
 - Processing.
 - Configuration.
 - Administration.
- Crash resiliency.

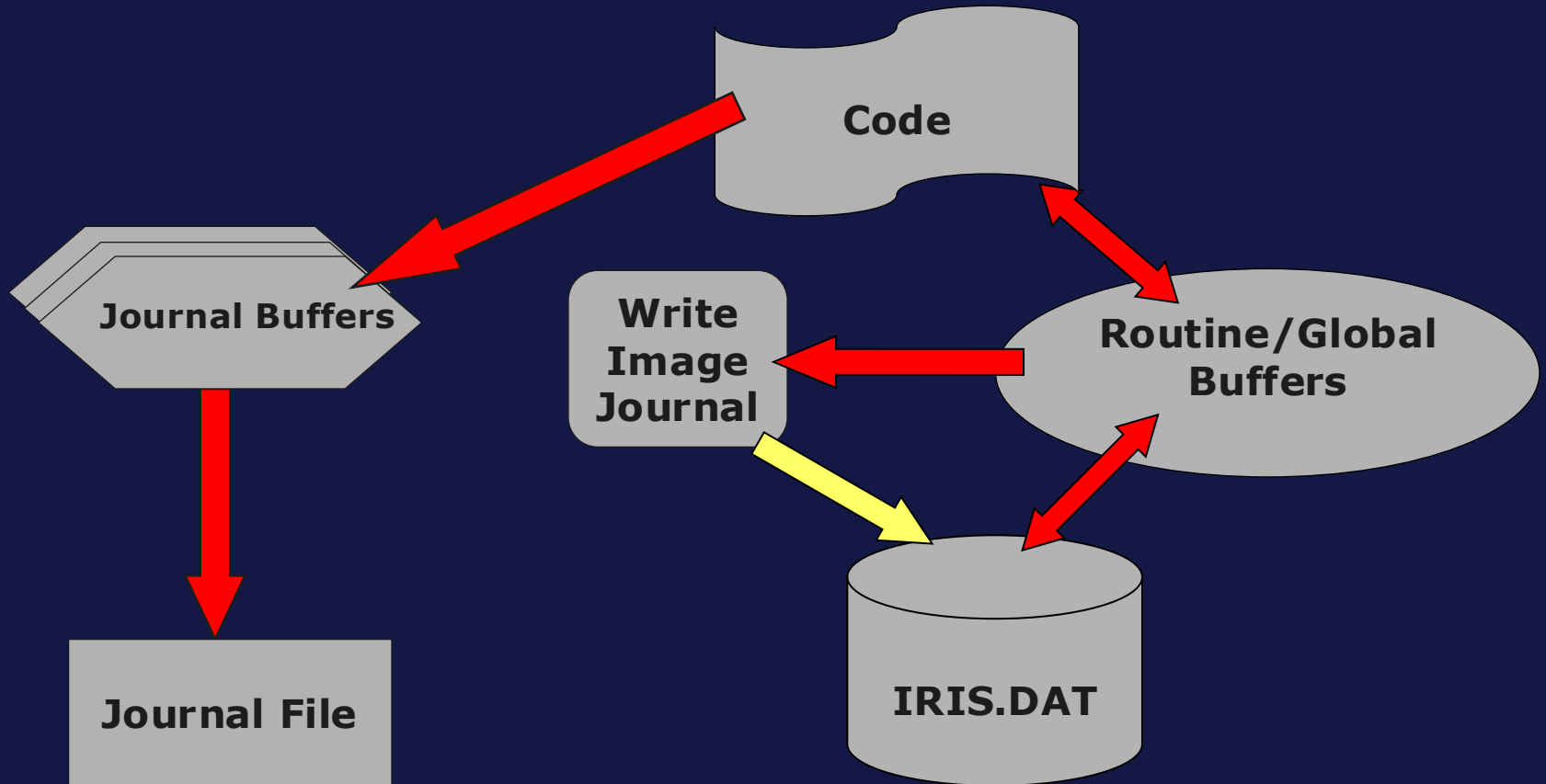


Types of Journaling

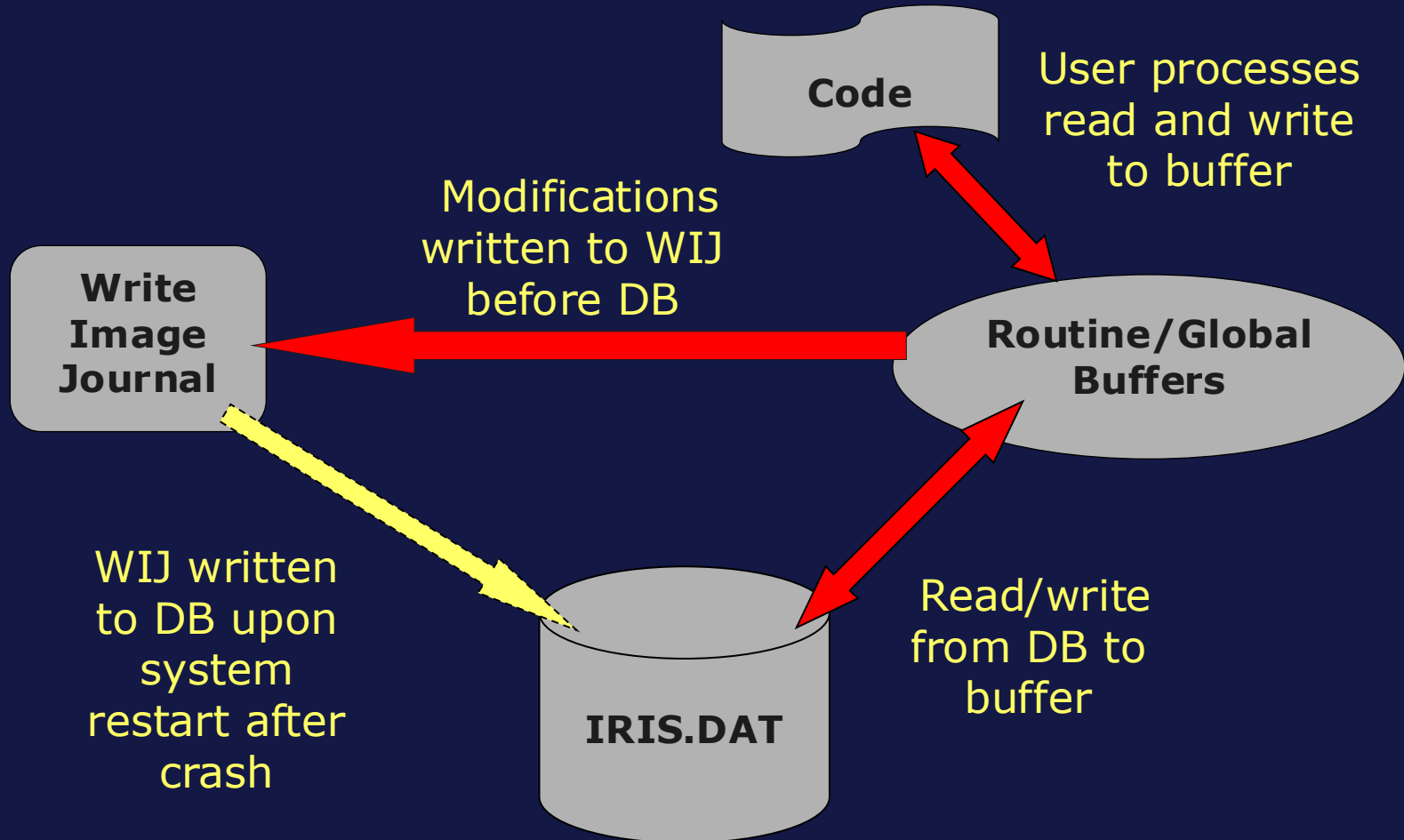
- Write Image Journal.
 - Record of planned database modifications before actual write into database.
 - Protects data integrity.
 - Prevents corruption.
- Journal.
 - Record of changes made to database.
 - Protects application integrity.
 - Prevents data loss or inconsistent data.



Types of Journaling (cont.)

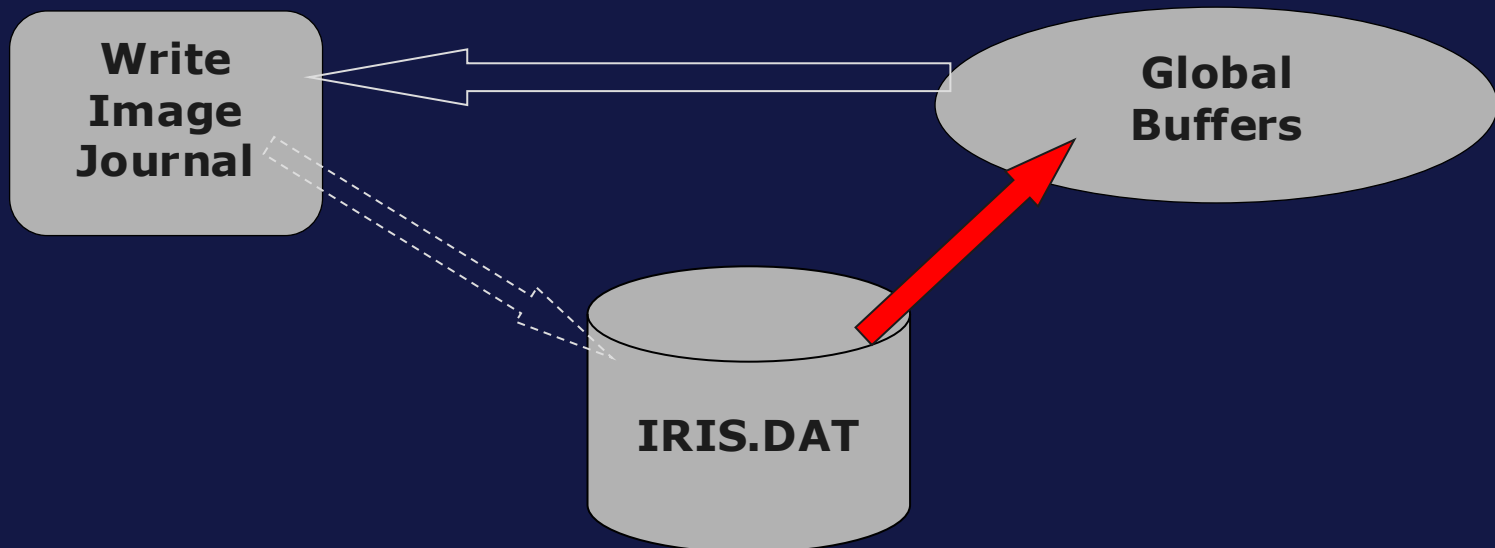


WIJ Overview



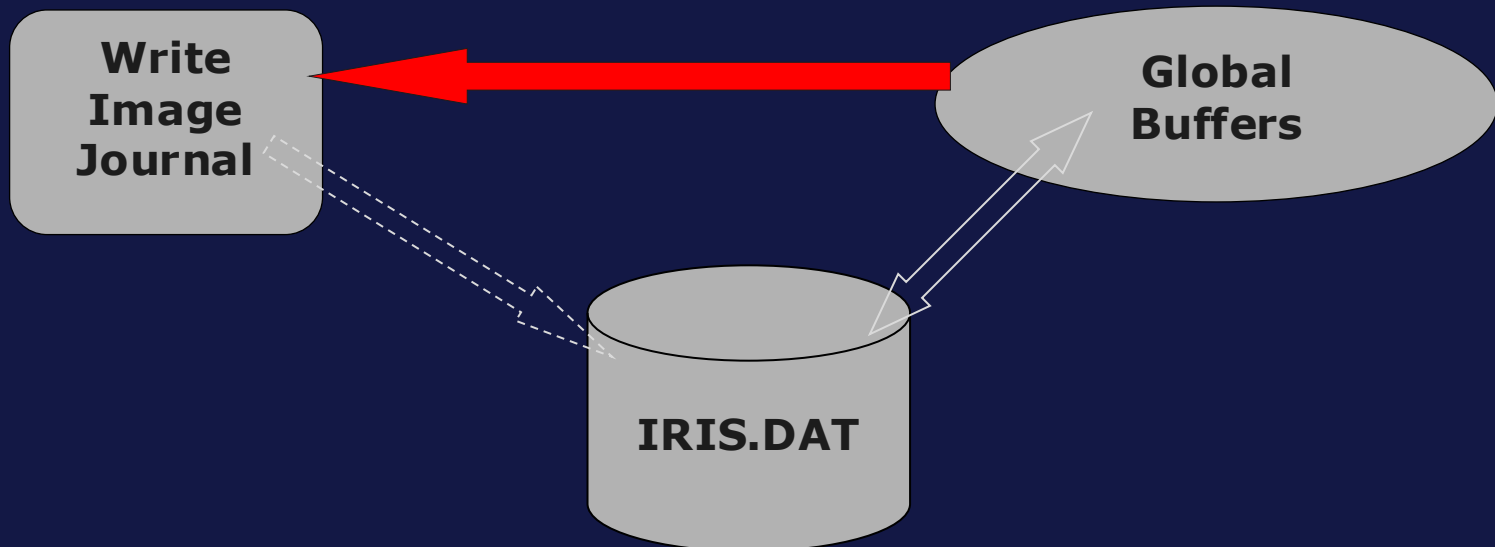
WIJ During Database Writes

- Process requests routine or global.
 - First,
 - System retrieves and places into buffer, if not already there.
 - Next,
 - Process accesses information in buffer.



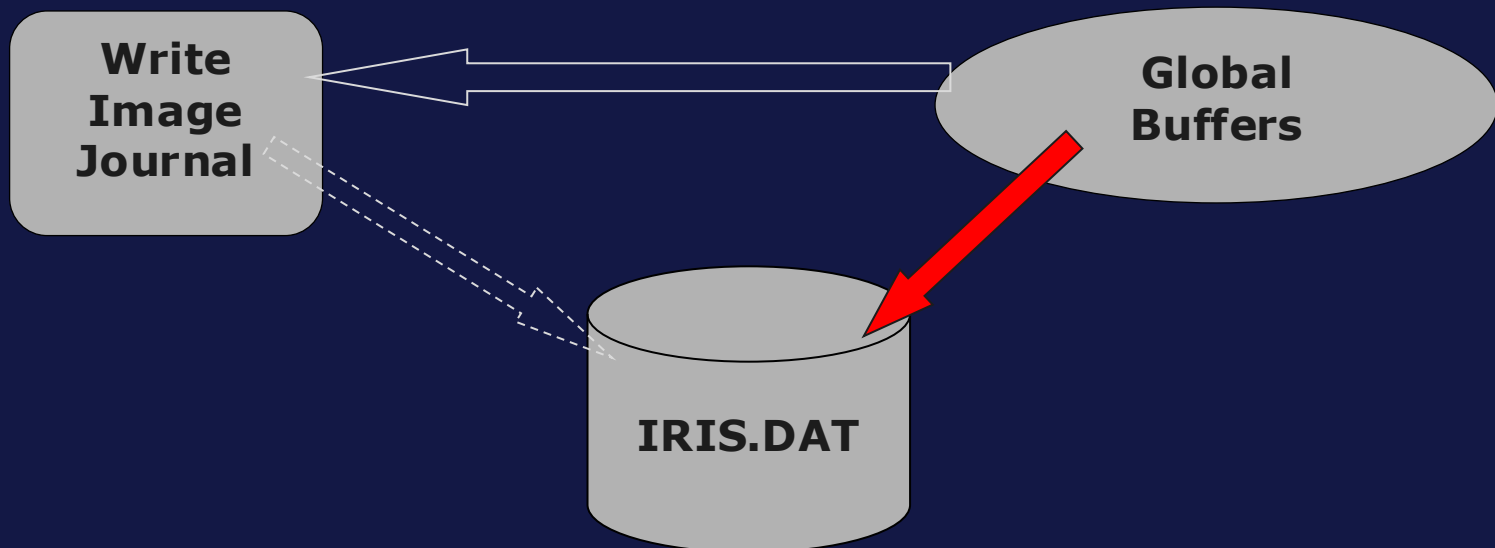
WIJ During Database Writes (cont.)

- Process changes routine/global.
 - First,
 - Write daemon wakes.
 - Write daemons writes changed buffers into WIJ.
 - Then, flag is set to mark WIJ active.



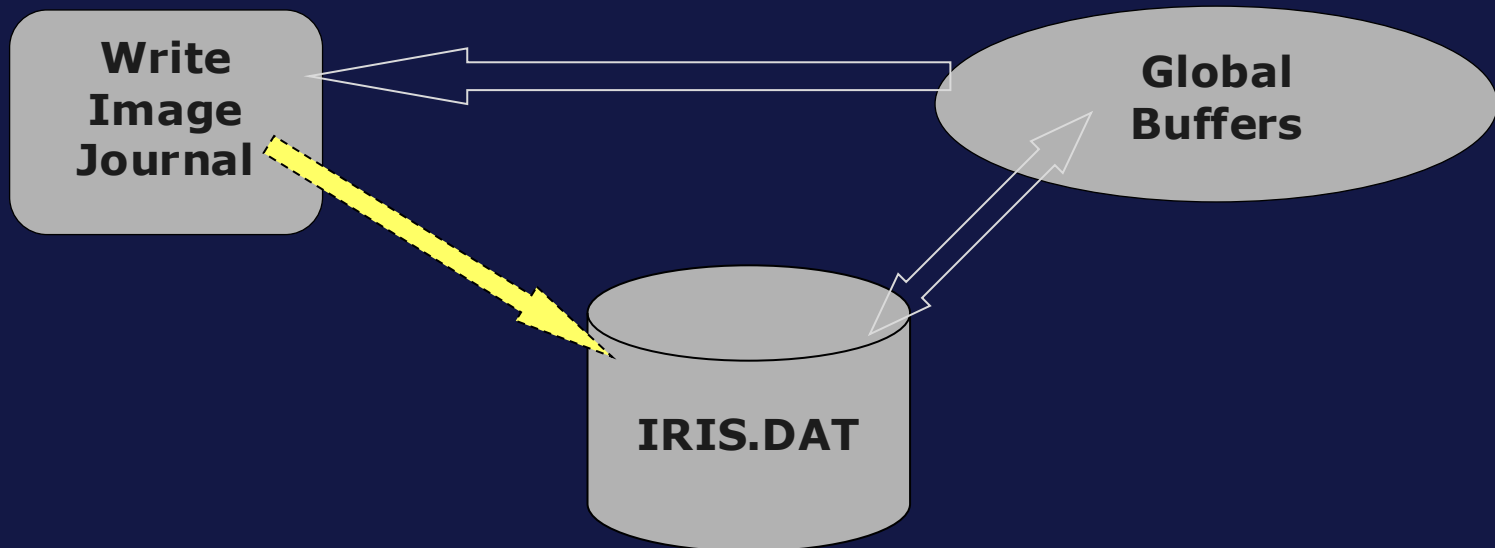
WIJ During Database Writes (cont.)

- Process changes routine/global (cont.).
 - Next,
 - Write daemon writes information into database from buffers.
 - Then, flag is reset to mark WIJ inactive.
 - Write daemon sleeps.



WIJ During Recovery

- If abnormal shutdown detected upon start, system automatically rewrites any updates in WIJ to database.
 - WIJ is rolled forward.



WIJ During Recovery (cont.)

- If write daemon failed during WIJ write phase, WIJ is **not** marked active.
 - WIJ information ignored.
 - Journal will be used to restore information.



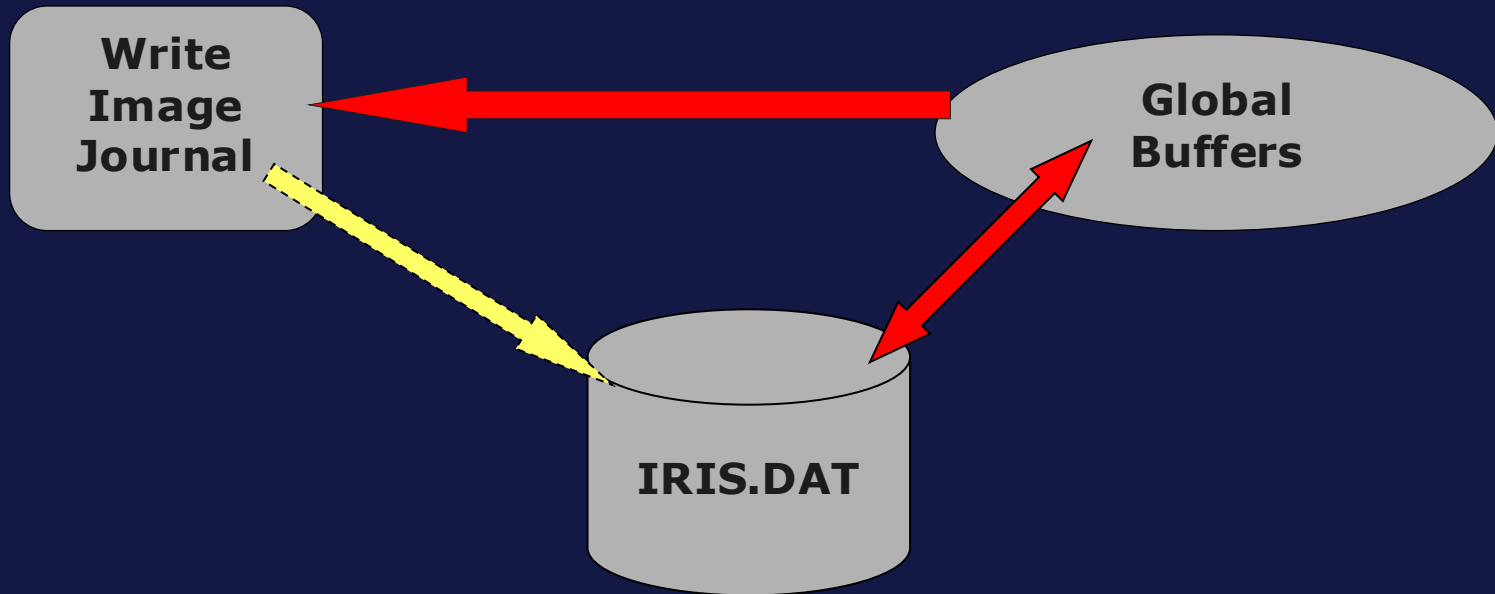
Write Image Journal File

- IRIS.WIJ.
- Expands as necessary.
 - Need enough file system space for WIJ to potentially expand to size of global cache.
- If WIJ file does not exist and cannot be created during startup:
 - System will not start.
- Do not delete WIJ.
 - Exception: When restarting after restoring backup IRIS.DATs (Backup module).



WIJ Processing

- WIJ processing automatically enabled on system start.



WIJ Directory

- System Administration > Configuration > System Configuration > Journal Settings.
 - Defaults to *InstallDirectory/mgr* directory.
 - Default path appropriate.

Primary journal directory Browse...
Required.

Secondary journal directory Browse...

Start new journal file every (MB)
Required.

Journal file prefix

When to purge journal files

After this many days
Required. (0 - 100)

After this many successive backups
Required. (0 - 10)

If both numbers are defined as nonzero, purging occurs when either condition is met.

Freeze on error ☐

Journal Web session ☐

Write image journal directory Browse...
Activating a WIJ directory change on cluster members requires restart.

Target size for the wij (MB) (0=not set)
Required.



Journaling Uses

- Record of changes made to data.
- Supports restoring data from **backup**.
 - Referred to as roll forward recovery.
- Supports **transaction** processing.
- Provides **crash resiliency** with WII.
 - Allows recovery of journal records and rollback.
- Required for **Mirroring**.



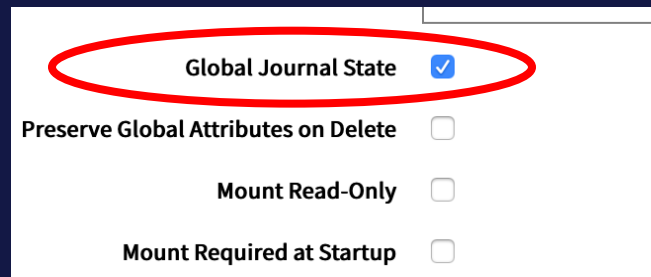
Journaling Database

- No need to journal all databases
 - Library databases (IRISLIB) – readonly, no change
 - Temp database (IRISTEMP) – no need to recover
 - Any other databases no need to recover
- Below system databases need to be journaled
 - IRISADUIT
 - IRISYS
 - IRISSECURITY

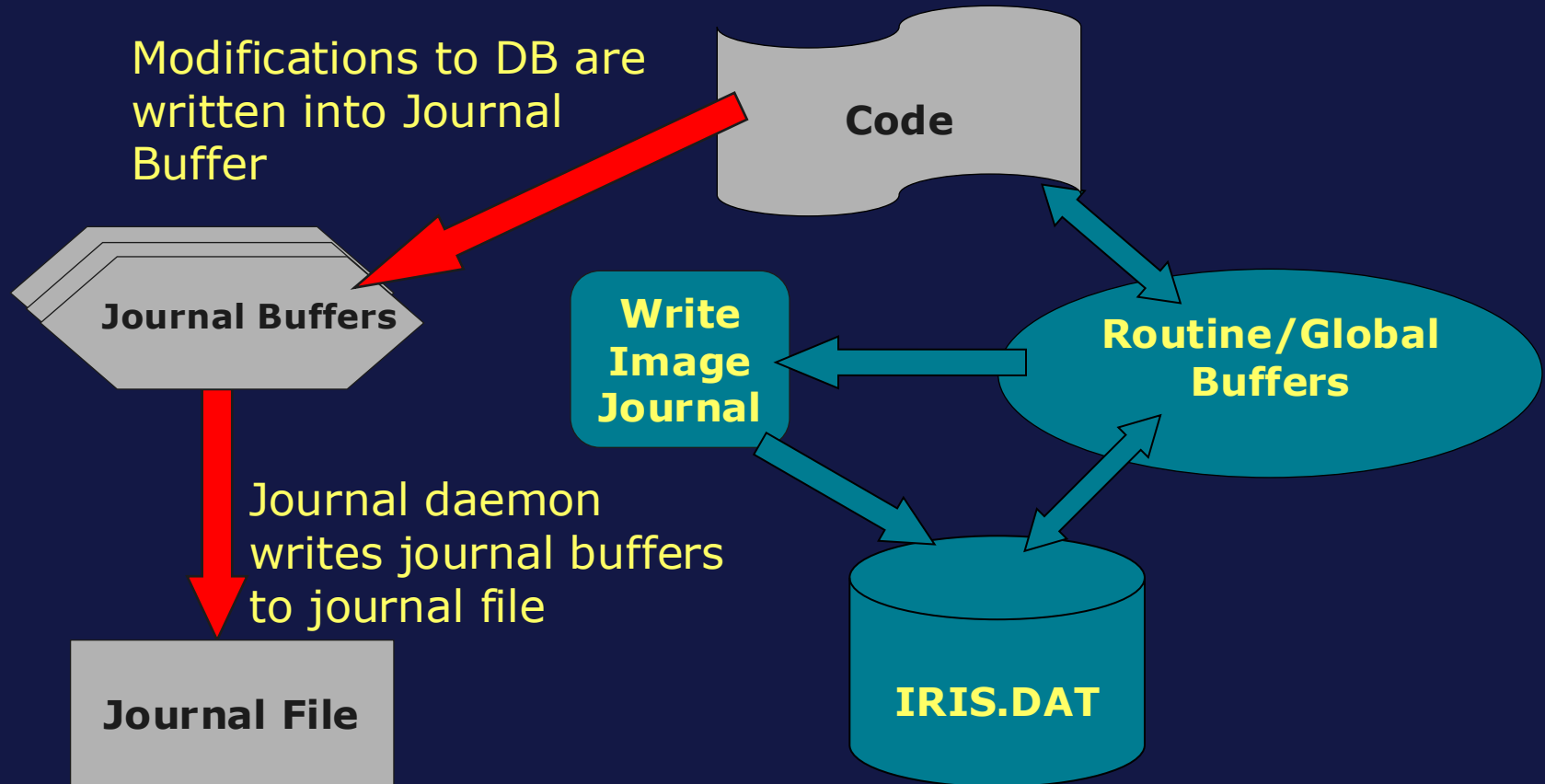


Global Journal State

- System Administration > Configuration > System Configuration > Local Databases > Select database.
 - Checked (default, recommended).
 - Journals all globals in database.
 - Required for transactional consistency after crash.
 - Unchecked.
 - Only journals globals in a transaction.
 - Default for IRISLOCALDATA, IRISLIB and ENSLIB.
 - IRISTEMP never journaled.



Journaling Flow



Configuring Journaling

System > Configuration > Journal Settings

Primary journal directory	C:\PrimaryJournals	Browse...
	Required.	
Secondary journal directory	D:\SecondaryJournals	Browse...
Start new journal file every (MB)	1024	
	Required.	
Journal file prefix		
When to purge journal files	After this many days	2
		Required. (0 - 100)

- Primary journal directory
 - Default is *InstallDirectory*/mgr/journal.
 - Move to isolated disk.
- Secondary journal directory
 - Alternate directory used when primary fills.
 - Continues to use secondary unless manually changed.



Configuring Journaling (cont.)

System > Configuration > Journal Settings

When to purge journal files

After this many days
Required. (0 - 100)

After this many successive backups
Required. (0 - 10)

If both numbers are defined as nonzero, purging occurs when either condition is met.

Freeze on error ☐

Journal Web session ☐

Write image journal directory
Activating a WIJ directory change on cluster members requires restart.

Target size for the wij (MB) (0=not set)
Required.

■ FreezeOnError

- Controls behavior when error occurs in writing to journal.
- Recommend set to freeze.
 - Default is “No”.
- If do not freeze, need to do full backup after error is fixed since cannot use journal file to recover database.



Configuring Journaling (cont.)

System > Configuration > Journal Settings

Primary journal directory C:\PrimaryJournals Browse...
Required.

Secondary journal directory D:\SecondaryJournals Browse...

Start new journal file every (MB) 1024
Required.

Journal file prefix

Default

- Start new journal file every (MB)
 - Switches to new file when max size reached.
- Journal File Prefix
 - Adds prefix to journal file name.
 - Used to more easily identify journals from multiple servers.



Journal File

- *Date.Number* (for example: 20241126.001).
 - Compressed files end in "z".
- Contains:
 - Sets and kills for globals.
 - Transaction begins and commits.

Offset	Time	Proc	Type	InTxn	Global	Database
135140	2014-09-15 16:22:32	11596	KILL	No	^SYS("CLMANAGER")	c:\intersystems
135188	2014-09-15 16:22:32	11596	SET	No	^SYS("CLMANAGER")	c:\intersystems
135240	2014-09-15 16:22:32	11596	SET	No	^SYS("CLMANAGER",1)	c:\intersystems
135296	2014-09-15 16:22:32	11596	SET	No	^SYS("CLMANAGER",1,"started")	c:\intersystems
135540	2014-09-15 16:22:32	12080	SET	No	^SYS("Security","UsersD","cspsystem")	c:\intersystems
135612	2014-09-15 16:22:32	31272	SET	No	^MIRROR("NEWMIRROR","retry","list")	c:\intersystems
135680	2014-09-15 16:22:32	31272	SET	No	^MIRROR("NEWMIRROR","retry","list",1)	c:\intersystems
135820	2014-09-15 16:22:32	31272	SET	No	^MIRROR("NEWMIRROR","retry","list",1,"goto")	c:\intersystems
135896	2014-09-15 16:22:32	31272	SET	No	^MIRROR("NEWMIRROR","RetryReason")	c:\intersystems
135988	2014-09-15 16:22:32	31272	SET	No	^MIRROR("NEWMIRROR","Role")	c:\intersystems
136044	2014-09-15 16:22:32	31272	SET	No	^MIRROR("NEWMIRROR","Status","MGR")	c:\intersystems
136108	2014-09-15 16:22:32	26728	BeginTrans	Yes		
136124	2014-09-15 16:22:32	26728	SET	No	^ISC.Monitor.AlertD	c:\intersystems
136180	2014-09-15 16:22:32	26728	SET	Yes	^ISC.Monitor.AlertD(7)	c:\intersystems
136336	2014-09-15 16:22:32	26728	CommitTrans	Yes		

Time journal buffer starts filling.



Journal File Format

- In v2025.1+, journal file uses more compact format.
- **Cannot** use new format journal file on pre-v2025.1 systems.
- **Can** use pre-v2025.1 journal file on v2025.1+ systems.
 - Restoring from backup.
 - While upgrading mirror.



Managing Journal Files

System > Configuration > Journal Settings

Journal file prefix

When to purge journal files

After this many days

Required. (0 - 100)

After this many successive backups

Required. (0 - 10)

If both numbers are defined as nonzero, purging occurs when either condition is met.

Freeze on error



- Purge journal files automatically.
 - Based on days and/or successive successful backups.



Managing Journal Files (cont.)

- Enable journal compression.
 - Only compresses completed journal files.
 - System Administration > Configuration > System Configuration > Journal Settings.
 - Enabled by default in new installations.
 - Upgrading maintains current setting.
- Archive journal files. (v2023.3+)
 - Send inactive journal files to archive location.
 - Option to purge journal files after being copied.
 - If enabled, purging no longer based on days and/or successive backups.



Managing Journal Files (cont.)

- To manually purge journal files:
 - Do not delete journal files directly from OS.
 - %SYS>do ^JOURNAL
 - %SYS>do PPURGE^JRNUTIL
- Backup journal files at OS level.



Viewing and Searching Journal File

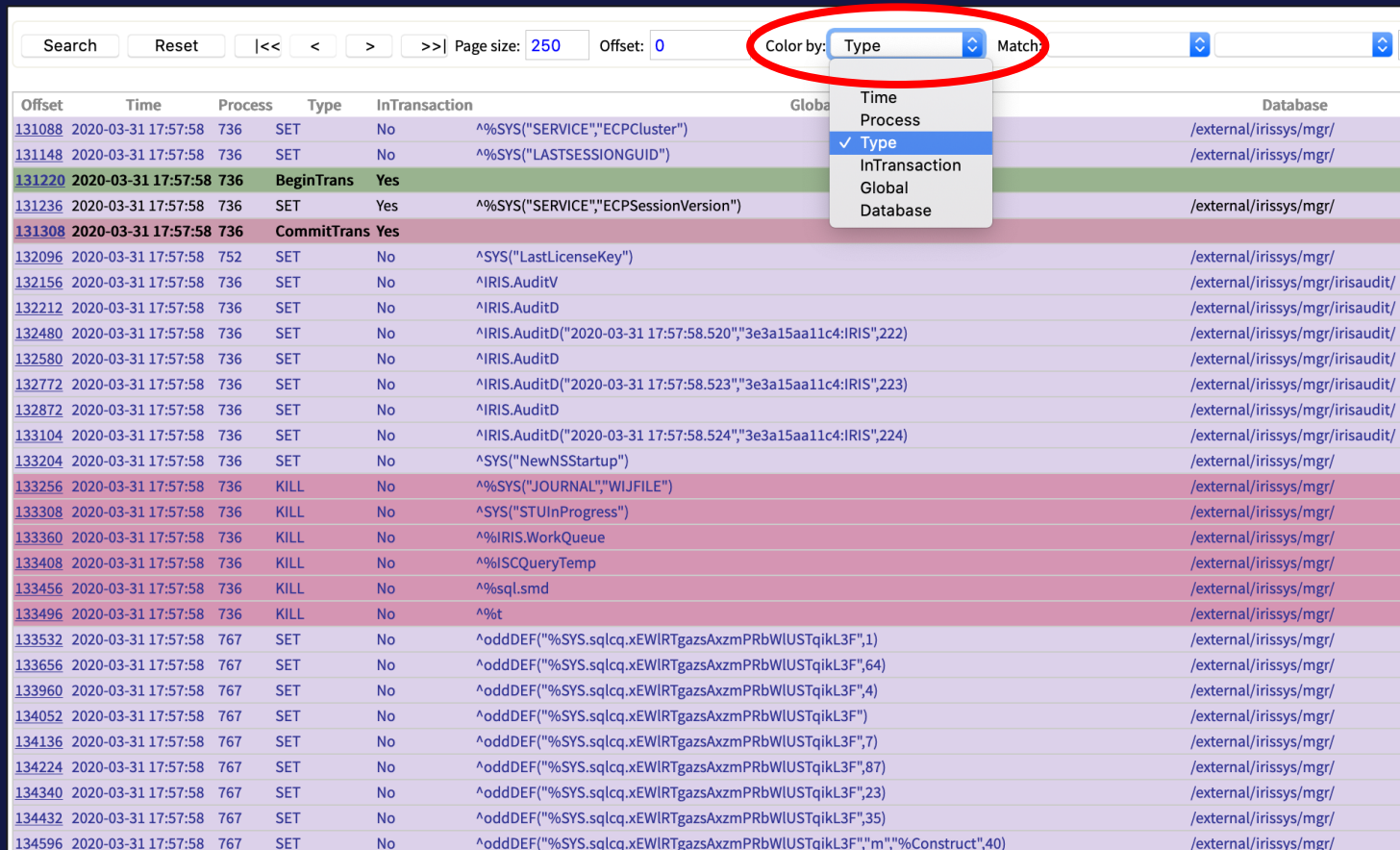
- System Operation > Journals > View.
 - List of files shows:
 - Size (compressed size for completed journals).
 - Data Size (size of journaled data).
- %SYS>do ^JOURNAL.
- %SYS>do ^JRNDUMP.
- Can search journal file by global, process ID, database, time, and others.

Name	Size	Creation Time	Reason	Data Size				
/Host/SYS/mgr/journal/20260310.002	192 KB	2026-03-10 21:57:51	<STARTUP>	192 KB	View	Profile	Integrity Check	Summary
/Host/SYS/mgr/journal/20260310.001	564 KB	2026-03-10 00:00:01	by task manager	4160 KB	View	Profile	Integrity Check	Summary
/Host/SYS/mgr/journal/20260309.001	160 KB	2026-03-09 00:42:27	by task manager	2560 KB	View	Profile	Integrity Check	Summary
/Host/SYS/mgr/journal/20260308.001	116 KB	2026-03-08 00:00:50	by task manager	1024 KB	View	Profile	Integrity Check	Summary



Viewing a Journal File (cont.)

- Can change coloring of journal entries.



Offset	Time	Process	Type	InTransaction	Global	Database
131088	2020-03-31 17:57:58	736	SET	No	^%SYS("SERVICE","ECPCluster")	/external/irissys/mgr/
131148	2020-03-31 17:57:58	736	SET	No	^%SYS("LASTSESSIONGUID")	/external/irissys/mgr/
131220	2020-03-31 17:57:58	736	BeginTrans	Yes		/external/irissys/mgr/
131236	2020-03-31 17:57:58	736	SET	Yes	^%SYS("SERVICE","ECPSessionVersion")	/external/irissys/mgr/
131308	2020-03-31 17:57:58	736	CommitTrans	Yes		/external/irissys/mgr/
132096	2020-03-31 17:57:58	752	SET	No	^SYS("LastLicenseKey")	/external/irissys/mgr/
132156	2020-03-31 17:57:58	736	SET	No	^IRIS.AuditV	/external/irissys/mgr/irisaudit/
132212	2020-03-31 17:57:58	736	SET	No	^IRIS.AuditD	/external/irissys/mgr/irisaudit/
132480	2020-03-31 17:57:58	736	SET	No	^IRIS.AuditD("2020-03-31 17:57:58.520","3e3a15aa11c4:IRIS",222)	/external/irissys/mgr/irisaudit/
132580	2020-03-31 17:57:58	736	SET	No	^IRIS.AuditD	/external/irissys/mgr/irisaudit/
132772	2020-03-31 17:57:58	736	SET	No	^IRIS.AuditD("2020-03-31 17:57:58.523","3e3a15aa11c4:IRIS",223)	/external/irissys/mgr/irisaudit/
132872	2020-03-31 17:57:58	736	SET	No	^IRIS.AuditD	/external/irissys/mgr/irisaudit/
133104	2020-03-31 17:57:58	736	SET	No	^IRIS.AuditD("2020-03-31 17:57:58.524","3e3a15aa11c4:IRIS",224)	/external/irissys/mgr/irisaudit/
133204	2020-03-31 17:57:58	736	SET	No	^SYS("NewNSStartup")	/external/irissys/mgr/
133256	2020-03-31 17:57:58	736	KILL	No	^%SYS("JOURNAL","WIJFILE")	/external/irissys/mgr/
133308	2020-03-31 17:57:58	736	KILL	No	^SYS("STUInProgress")	/external/irissys/mgr/
133360	2020-03-31 17:57:58	736	KILL	No	^%IRIS.WorkQueue	/external/irissys/mgr/
133408	2020-03-31 17:57:58	736	KILL	No	^%ISCQueryTemp	/external/irissys/mgr/
133456	2020-03-31 17:57:58	736	KILL	No	^%sql.smd	/external/irissys/mgr/
133496	2020-03-31 17:57:58	736	KILL	No	^%t	/external/irissys/mgr/
133532	2020-03-31 17:57:58	767	SET	No	^oddDEF("%SYS.sqlcq.xEWIRTgazsAxzmPRbWIUSTqikL3F",1)	/external/irissys/mgr/
133656	2020-03-31 17:57:58	767	SET	No	^oddDEF("%SYS.sqlcq.xEWIRTgazsAxzmPRbWIUSTqikL3F",64)	/external/irissys/mgr/
133960	2020-03-31 17:57:58	767	SET	No	^oddDEF("%SYS.sqlcq.xEWIRTgazsAxzmPRbWIUSTqikL3F",4)	/external/irissys/mgr/
134052	2020-03-31 17:57:58	767	SET	No	^oddDEF("%SYS.sqlcq.xEWIRTgazsAxzmPRbWIUSTqikL3F")	/external/irissys/mgr/
134136	2020-03-31 17:57:58	767	SET	No	^oddDEF("%SYS.sqlcq.xEWIRTgazsAxzmPRbWIUSTqikL3F",7)	/external/irissys/mgr/
134224	2020-03-31 17:57:58	767	SET	No	^oddDEF("%SYS.sqlcq.xEWIRTgazsAxzmPRbWIUSTqikL3F",87)	/external/irissys/mgr/
134340	2020-03-31 17:57:58	767	SET	No	^oddDEF("%SYS.sqlcq.xEWIRTgazsAxzmPRbWIUSTqikL3F",23)	/external/irissys/mgr/
134432	2020-03-31 17:57:58	767	SET	No	^oddDEF("%SYS.sqlcq.xEWIRTgazsAxzmPRbWIUSTqikL3F",35)	/external/irissys/mgr/
134596	2020-03-31 17:57:58	767	SET	No	^oddDEF("%SYS.sqlcq.xEWIRTgazsAxzmPRbWIUSTqikL3F","m","%Construct",40)	/external/irissys/mgr/



Journal Profile

- System Operation > Journals > Profile.

Sort by: ☒ Global ☐ Size Recalculate Cancel Ready

c:\intersystems\training\mgr\journal\20200331.003

Database	Global	Size (Bytes)
c:\intersystems\training\mgr\	%ISCQueryTemp	48
c:\intersystems\training\mgr\	%SYS	300
c:\intersystems\training\mgr\	%t	36
c:\intersystems\training\mgr\	SYS	5588
isaudit\	IRIS.AuditD	672
isaudit\		

Globals with entries in journal file

#of bytes of journal data for global



Switching a Journal File

- To switch journal file:
 - System Operation > Journals > Switch Journal.
 - %SYS>do ^JOURNAL.
 - %SYS>do ^JRNSWCH.
- Automatically switches at midnight.
- Backup automatically switches journal.
- Last entries from previous journal also written to new journal.
 - Up to 64KB.



Journal Restore

- Restore journals in order from oldest to most recent after system backup restored.
- Options to restore journal:
 - Part of backup restore process using %SYS>do ^BACKUP.
 - Prompts to restore.
 - %SYS>do ^JOURNAL.
 - %SYS>do ^JRNRESTO.
 - Use Journal.Restore class.
- Example:
 - Full backup file:
c:\iris\mgr\Backup\FULL\20240117_001.cbk
 - Journal Files:
c:\iris\mgr\journal\20240117.001
c:\iris\mgr\journal\20240117.002
c:\iris\mgr\journal\20240117.003



Encrypted Journal Files

- Journal files can be encrypted.
- Encrypt using %SYS>do ^JOURNAL.
- Refer to Encryption module for more information.

```
%SYS>do ^JOURNAL
```

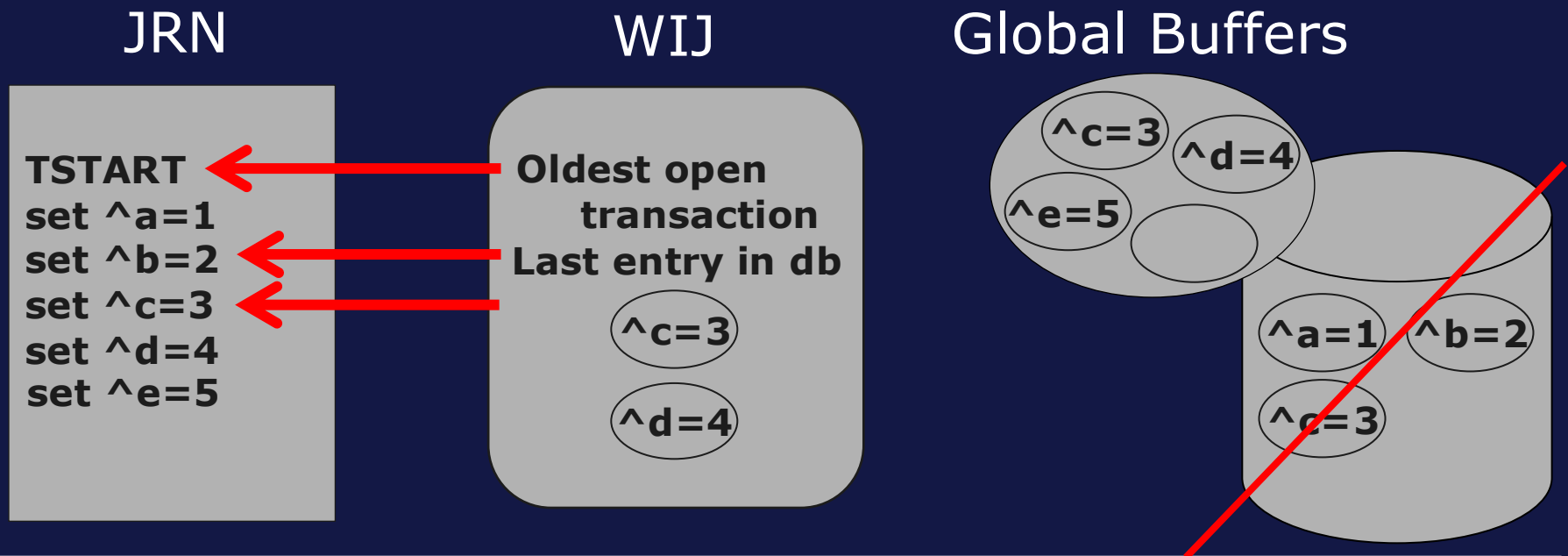
```
1) Begin Journaling (^JRNSTART)  
2) Stop Journaling (^JRNSTOP)  
3) Switch Journal File (^JRNSWTCH)  
4) Restore Globals From Journal (^JRNRESTO)  
5) Display Journal File (^JRNDUMP)  
6) Purge Journal Files (PURGE^JOURNAL)  
7) Edit Journal Properties (^JRNOPTS)  
8) Activate or Deactivate Journal Encryption (ENCRYPT^JOURNAL())  
9) Display Journal status (Status^JOURNAL)  
10) -not available-  
11) -not available-  
12) Journal catch-up for mirrored databases (MirrorCatchup^JRNRESTO)  
13) -not available-
```

```
Option?
```



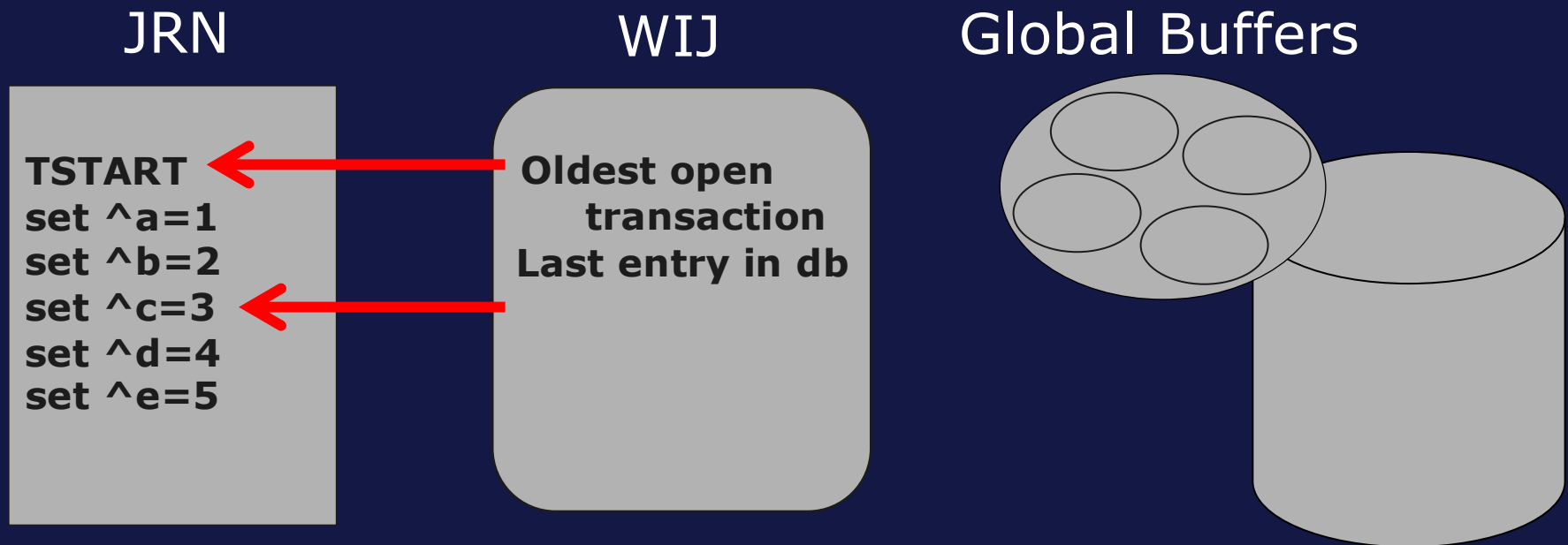
Crash Resiliency

- Besides blocks being written to disk, WIJ contains:
 - Pointer to last journal entry successfully written to database.
 - Address of oldest open transaction.



Crash Resiliency (cont.)

- Upon restart after crash, system automatically:
 - Rolls forward WIJ.
 - Rolls forward journal from pointer in WIJ.
 - Rolls back all open transactions from pointer in WIJ.
- Database never left in inconsistent state.



Suggested Readings

- *Data Integrity Guide.*
 - *Write Image Journaling and Recovery.*
 - *Journaling.*
- *Using ObjectScript > Transaction Processing.*



Summary

- What are the key points for this module?

